

"I store some passwords in WhatsApp": Understanding User Experiences and Usability Challenges of Password Managers in Virtual Reality

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Abstract

As virtual reality (VR) moves beyond entertainment into everyday use, secure user authentication becomes critical, yet passwords remain the dominant mechanism. Entering complex passwords with controllers is slow, error-prone, and vulnerable to observation, making password managers (PMs) a promising solution, whose usability for VR remains underexplored. As VR users start to rely on new VR-native or traditional managers, we lack understanding on how they select, trust, and interact with these tools. To bridge this gap, we conducted the first interview-based study with 11 users of PMs in VR. Our findings show that while PMs enhance password usability in VR, they do not fully meet VR users' needs, leading them to rely on their own risky security practices. Based on the findings, we recommend that future VR PMs consider tool familiarity, incorporate VR-native, biometric-enabled password management, and provide simplified, interactive security feedback to further enhance usability, favoring trust and secure practices.

CCS Concepts

• **Security and privacy** → **Usability in security and privacy**; **Authentication**.

Keywords

Virtual Reality, Password Managers, User Study

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1 Introduction

Virtual reality (VR) moves beyond entertainment into everyday use, making secure user authentication critical. As recent studies indicate [12, 29], passwords are the dominant mechanism, despite known usability and security challenges. It is complex to enter

passwords with virtual keyboards using VR controllers or free hand interactions, and these input methods are vulnerable to observation attacks [7, 14, 32].

To address these challenges, password managers (PMs) are a well-established approach [2, 18, 21] that enables automated login through the autofill feature, as well as security-enhancing functionalities like strong password generation and security audits. Recent releases of VR-specific password managers [15], along with increasing adoption and interest in using them [3, 12, 25], underscore the growing relevance of these tools within the VR ecosystem. While recent research by Düzgün et al. [5] and Stephenson et al. [29] suggest PMs as a recommended XR authentication solution, their suitability remains under-explored beyond the expert-based usability review in [12], where findings reveal that some VR-deployed solutions are less functional than traditional managers and lack critical features like autofill (across apps), password editing, and secure VR-friendly configuration. Yet, little is known about the actual usability challenges and experiences of users of PMs in VR. We conduct the first study with this population, to research:

RQ1 What are users' experiences and usability challenges with password managers in VR?

RQ2 How can PMs in VR be improved based on user suggestions?

Our interviews with 11 participants who use a password manager in VR bring new insights relevant for future VR PM design. First, familiarity with the VR device increased trust in PM security, while some participants perceived using PMs in VR as less trustworthy due to the risk of device theft. Participants also noted that children should be considered for different security needs, and that onboardings can help users become familiar with the PM and its interface in VR. Furthermore, participants still rely on their own potentially insecure password practices (like storing passwords in WhatsApp or manually creating passwords), even when password managers offer supporting features, because existing solutions do not fully meet their needs in VR environments. Drawing on our findings, we provide recommendations for improving VR password managers.

2 Related Work

Research has investigated and introduced numerous authentication schemes for VR, including knowledge-based, behavioral-based, and biometric-based schemes [7, 14, 16, 17, 22, 30]. Despite these innovations, studies show that passwords still remain the dominant authentication method in VR [12, 29]. Stephenson et al. [29] found



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that VR developers mainly use traditional passwords in their VR applications because they are easy to implement; however, passwords are recognized as unsuitable for VR due to their usability and security challenges. Düzgün et al. [5] conducted a series of workshops with security and HCI experts to collect 123 concept ideas for authentication in extended reality. Their findings indicate that achieving a balance between security, usability, and privacy may be possible through the use of XR password managers, recommending future research to explore their compatibility, usability, and security aspects. These insights emphasize the relevance of our study on password management in VR environments.

While research on password managers across desktops, mobile devices, and educational settings has deepened understanding of user motivations, adoption factors, and usability challenges [2, 18, 21, 24, 26, 27], studies examining their use and usability in virtual reality remain scarce. The closest related work, by Kablo et al. [12], surveyed 126 VR users on their password practices, finding out that many users experience password entry as cumbersome, often resorting to weaker passwords, aligning with [29], and most showed interest in adopting PMs in VR. Their expert analysis (91 expert walkthroughs of seven PMs across VR-native, browser-integrated, and web-based tools), showed that PMs in VR suffer from poor integration, missing features, and unclear security settings. However, no research has gone beyond expert evaluations to study first-hand user experiences; we address this gap.

3 Interview Study Methodology

Procedure. The interview consisted of five sections, using a script inspired by and adapted from previous work studying the usability of PM credential audit interfaces [11]. After obtaining consent for audio recording, we asked general questions about participants' use of VR and how they selected their password manager. The second block examined VR input methods to understand their potential role in later-reported usability issues. We then asked questions about the adoption and experience of the most used features inside a PM (autofill and password generation) [9, 18, 20, 21]. We also included questions about security reports to investigate how users interact with them in immersive versus conventional settings [10, 11]. In the third block, we discussed the overall experience in more depth, encountered usability challenges, trust perceptions, and comparison about experiences between VR and non-VR contexts, which allowed us to explore the synchronization feature, one of the most valued aspects of password managers [18]. The fourth block explored participants' suggestions for improvements, and the final block addressed intentions to continue using the PM in VR.

Data Analysis. Interview audio recordings were transcribed locally, anonymized, and analyzed through inductive coding [19]. One researcher created the codebook and coded the interviews. A second researcher coded the interviews independently using the codebook, achieving substantial agreement (Brennan and Prediger's Kappa [4] ≥ 0.7 [13]).

Study Setup. We recruited participants through Prolific [23], targeting interview-willing English-speakers who reported using a VR headset at least once per month and were located in the UK, the US, Germany, or France — countries that represent large VR markets worldwide [28] and have a good Prolific participant base.

Then, we screened participants who currently used PMs in VR in a pre-interview survey that also gathered contextual information on participants' VR usage patterns, number of accounts commonly accessed in VR, and demographics. Both survey and interview were piloted (with 5 and 1 participants, respectively), leading to no changes in the final instruments. The survey was run in July-August 2025 via LimeSurvey [8], and the interviews took place in Zoom [33] between August-October 2025. Both tools were hosted on university servers, compliant with the GDPR [6]. The survey averaged 3 minutes, and the interviews 23 minutes, with a compensation of 1.40 GBP and 9 GBP, respectively. All participants were over 18 and gave informed consent. Interviews were audio-only, and we never asked participants to share any personally identifiable information, passwords, or other sensitive data. The study is IRB-approved and all materials are included in the supplementary materials.

4 Results

We surveyed 300 participants and obtained 281 complete responses, of which 49 met the eligibility criteria and were invited for the interview. We report the results from the final 11 participants that completed the interview.

4.1 RQ1 - User Experience and Usability Challenges with PMs in VR

The interviewees' usage patterns about PMs, VR, and accounts are summarized in Table 1, and their demographics in Table 2. The sample covered a broad range of PMs and VR devices (5 each). All participants had between 1-5 password-protected VR accounts, with all except one accessing some or all of these passwords per VR session. LastPass was the most common PM (used by 6), accessed either through its VR app or a browser extension. Most interviewees were male; about half were aged 25-34 and held a bachelor's degree.

All participants reported using VR mainly for gaming. Other reported uses were entertainment purposes such as streaming (7), socializing and communication (6), productivity or work (4), banking (3), online shopping (2), and fitness (1). Participants confirmed that they use passwords for these activities. Regarding PM adoption in VR, the most common motivation was security and privacy (9), followed by convenience (7).

4.1.1 Usability and Performance Issues. In terms of usability and interface design, participants praised the PM's interface for being user-friendly, and noted that using a PM felt efficient, seamless, and makes tasks faster (6). However, participants also reported difficulties. The main issue concerned the virtual keyboard: participants found it challenging to enter their primary password, as it often led to typing errors (4): *The master password for my Bitwarden is quite complicated. It's a bit of a hassle to have to type that in using the virtual keyboard (P6).* Further, accessing a third-party PM on VR, which is only available as a web app, needed an additional setup step as they first needed to launch the browser, mentioned by two.

The lack of the autofill feature of some PMs resulted in complaints about having to rely on copy-and-paste, across other apps or websites for two participants, as it requires multiple steps to log in: *The autofill [...] doesn't work on the Steam app or different games.[...] I have to switch to the browser and manually find what account I'm looking for, copy the password, switch back to the game,*

Table 1: Summary table containing interviewees' VR, PM, and accounts usage patterns

ID	Password Manager	VR Device	Monthly VR Usage	VR Accounts	Acc. in 1 session	PM outside VR
P1	Meta Quest PM	Meta Quest 3	1-5 times	1-5	~ All	Different
P2	1Password Web	Oculus Rift	1-5 times	1-5	~ 25%	Same
P3	LastPass VR App	Meta Quest 3	>15 times	1-5	~ 75%	Different
P4	Meta Quest PM	Meta Quest Pro	11-15 times	1-5	~ half	Different
P5	LastPass Extension	Meta Quest 3	1-5 times	1-5	~ half	Same
P6	Bitwarden Extension	Apple Vision Pro	1-5 times	1-5	~ 25%	Same
P7	LastPass Extension	Meta Quest 2	1-5 times	1-5	~ 25%	Same
P8	LastPass VR App	Meta Quest 2	11-15 times	1-5	None	Same
P9	Meta Quest PM	Meta Quest 2	6-10 times	1-5	~ half	Different
P10	LastPass Extension	Meta Quest 3	1-5 times	1-5	~ half	Same
P11	LastPass VR App	Meta Quest 2	1-5 times	1-5	~ All	Same

paste it in, and switch back to copy the email (P6). One participant noted that their PM lacks an autofill feature, but nevertheless chose to use it in VR because it synchronizes passwords across their devices. Regarding password generation, one user chooses not to use it because they prefer creating their own memorable passwords, even though the feature is visible during account creation. Another one was not aware of the password generation feature and hence does not use it explaining that they know how to create complex passwords themselves and find manually created ones easier to manage. Regarding security reports, one participant explained that they ignore the reports to keep their interactions in VR simple and avoid unnecessary complexity.

4.1.2 Differences to PMs in Other Settings. Seven participants stated to use the same PM outside VR as well. Reasons are cross-device synchronization of the passwords to access those which were saved before on other devices. Those were all users of third-party PMs (LastPass, 1Password, Bitwarden). The remaining four participants stated that they use a different PM outside of VR, because their PM is not available on VR such as Google Chrome PM.

Usability Differences. The VR interaction is more challenging, as input methods are different, mentioned by two. Two others stated they are more familiar with non-VR devices, in particular seeing the PM stretched out in 3D is irritating, or that unfamiliarity leads to slow typing. In general, six participants stated that the PM VR version is slower and less reliable. On the other hand, one person enjoys interacting with the PM in VR because it feels immersive.

Features Differences. Since readability is better on non-VR devices, one person prefers to interact with security reports outside of VR. In addition, another participant mentioned that they engage with security reports less frequently in VR compared to other settings because they use the VR headset only a few times per week and share it with family members. Regarding storing and accessing passwords in VR, one participant uses an alternative way in addition to the PM: *I've created a WhatsApp group just with myself. And there are some passwords that I store because it makes ease of use. [...] the copy and paste feature works really well, which sometimes I find LastPass does really struggle with* (P7).

Trust. Coming to trust perceptions, seven participants said they have equal trust in their PMs inside and outside of VR in terms of perceived security, three of them use a different PM outside VR. Reasons for equal trust are believing that the same security backend

protocols (e.g., encryption) is used across devices, self-hosting their own servers, or perceiving the password manager as trustworthy because it is natively integrated into the device's browser. Two participants expressed lower trust in the VR password manager, though using the PM outside VR, citing unfamiliarity with the device and concerns that it could be more easily stolen. Interestingly, one participant expressed higher trust in their VR PM, explaining that their laptop PM frequently displays breach notifications, which makes them feel unsafe, while in VR the absence of such alerts gives an impression of greater trust.

4.2 RQ2 - Improvements

UI, Usability, and Interaction. To improve user interaction, five participants suggested including voice commands for cancelling subscriptions, accessing FAQs, navigating to updating passwords, and selecting options without relying on controller input. One person wished for haptic feedback when a password is copied to make sure they selected it correctly. Eight participants suggested making the UI more VR friendly, e.g., using larger buttons and simpler layouts.

Features. Participants wanted a smoother password management experience in VR, including a native PM for easy setup (3) and autofill across apps (3), more reliable automatic filling of passwords (2), cross-device synchronization (1), and an autosave feature for unstored credentials across apps (1). Regarding security reports, one participant wished for regular risk assessments, such as weekly evaluations, both within and outside VR. Another participant specifically mentioned having an alert feature available within VR while another individual suggested the integration of notifications about OS and browser vulnerabilities. Further, other suggestions included tutorials on how to secure passwords (3). One described this as an onboarding guide for using the PM itself, while the others suggested a tutorial on general password management in VR.

Perceived Security. Because of the copy-pasting reliance, participants outlined the security risks of copying passwords into the clipboard (2). One of them suggested an auto delete of the content after a few seconds, so that other device users cannot access it, while the other proposed giving users control over clipboard data or having an encrypted clipboard that can only be accessed by the PM. One participant expressed concern about their child and suggested exploring ways to better support children, whose security needs might be higher. Another participant wished for security

Table 2: Interviewee demographics

		<i>n</i> = 11	Perc.
Gen.	Male	9	82%
	Female	2	18%
Age	35-44	2	18%
	25-34	5	45%
	45-54	2	18%
	18-24	2	18%
Educ.	Master's degr.	4	36%
	Bachelor's degr.	5	45%
	Associate degr.	2	18%
IT	No	5	45%
	Yes	5	45%
	No answer	1	9%

education that teaches users about the importance of changing breached passwords and keeping their data secure.

Alternative Authentication Methods. Several participants suggested using biometrics as an alternative authentication method (3), with some proposing them as replacements for primary password entry (2). Voice recognition was the most frequently mentioned option (2), while others referred to eye recognition, facial recognition, passkeys, and security questions as possible replacement for passwords (1 each).

Continued Usage. Finally, most participants intended to continue using their PM in VR for its availability, convenience, and security (9). Three participants anticipated increased use as their headset usage expands beyond gaming, while two indicated that their use would increase as PM features improve and VR applications become easier to navigate: *Especially if VR starts becoming a bigger part of work and study, I will see myself using it more in VR* (P5). One person might switch out of curiosity if an innovative solution emerged, another due to past LastPass data breaches, and one considered switching from Meta Quest PM to LastPass for better features.

5 Discussion

In our study, 37% of the active VR users we screened reported using PMs within VR. Although this indicates strong demand for such tools, our interviews revealed that current VR password management solutions still fall short of users' needs. Despite these shortcomings, many users continue to rely on them due to the lack of better alternatives. Among the rest of the participants who did not use PMs in VR, approximately half said they simply memorized their passwords, about one-third used a password manager outside of VR, and roughly 20% stored their passwords in digital or physical notes. Each of these alternatives presents issues: notebooks pose a security risk because they can be lost or stolen, memorizing passwords increases mental load and typically results in weaker passwords, and using PMs outside VR forces users to look up their credentials on another device, adding physical effort. Improving PMs for VR could encourage more users to adopt them or increase the frequency of use among those already relying on such tools, as two of our interview participants explicitly noted. In the following, we offer recommendations for VR platform and PM designers, and developers to improve PMs in VR.

Recommendations. Users may benefit when password managers function consistently across platforms, as familiarity can enhance trust and reduce the learning curve. However, this does not mean that PMs should simply replicate non-VR designs in VR environments without considering VR-specific needs. This point is supported by one participant, who chose to use a browser-based password manager in VR because they were already familiar with it and had positive prior experiences outside of VR. The PM should provide a seamless, VR-friendly interface with smooth interactions, particularly for entering the primary password until password-less methods, such as biometric options, become standard. Additionally, integrating VR-specific features such as voice commands and haptic feedback can further enhance user interaction, as well as a native PM with proper main features, as presented in 4.2, ideally integrated into the VR operating system, which could also increase security and privacy trust. Apple's Vision Pro uses Optic ID [1]

to identify users through iris scanning, enabling unlocking and password autofill. It works with iCloud Keychain, a system-level PM that stores and synchronizes credentials across apps and Apple devices. This combination seems promising for future authentication and password management in mainstream VR devices such as the Meta Quest series, which currently lack comparable built-in solutions.

Security reports are currently not widely adopted in VR due to their complexity, which users in previous work also found overwhelming [11]. One recommendation, as revealed by our findings and mentioned in [12] is to integrate these reports with a simplified interface and more intuitive functionality could encourage adoption, also in VR. Since the immersive experience is opening new opportunities, an idea could be to gamify those reports. As seen in 4.1.2, the relationship between trusting the PM and credential breach communication warrants further study as well.

To further reduce user overload, managers could include tutorials to guide VR users in PM usage. VR users would also benefit from educational views that explain how to securely store passwords and what management options are available. Such features could additionally support specific user groups, such as children, who frequently use VR headsets for gaming and may have different security needs than adults [31]. This point was raised by one participant. As participants noted, VR headsets are often shared devices that require specific protections for multi-user data. We recommend implementing separate profiles, echoing expert-based research [12], and clipboard protection mechanisms. Further comparisons with prior work reveal similar usability and security challenges. Kablo et al. [12] and Stephenson et al. [29] likewise observed slow, error-prone password entry due to virtual keyboards and controller tracking. Autofill was considered essential, while reliance on copy-paste across apps and extra setup for web-based PMs were viewed as inconvenient and risky. Consistent with earlier findings, our participants also tended to prefer simpler, manually created passwords over generated ones.

Limitations and Future Work. Our findings are based on participants' self-reports of their VR experiences as recalled from memory; behaviors could not be directly examined or measured. It is also possible that participants under-reported perceived insecure behaviors due to social desirability bias. Although the interviews yielded rich qualitative data, the limited sample restricts generalizability. While the screened participant pool was relatively gender balanced, the interview sample was predominantly male. This imbalance may reflect the demographics of the current early-adopter VR community. Future research work could focus on controlled laboratory studies to directly observe real-time behavior on how users perform logins, navigate between VR windows, manage copy-paste workflows, or create new credentials inside VR. This would enable the collection of objective performance data and provide a finer-grained understanding of efficiency, error patterns, and interaction challenges.

Use of Large Language Models. We used OpenAI's GPT-5 model via OpenAI's API using Open WebUI for editing and improving the writing for this publication. All findings are our own, and were informed by our background, the data of this study, and the cited literature.

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